KOTAANA

How the app thinks

The workout algorithm, the nutrition algorithm, the dashboard maths and every API that connects them — explained step by step, with diagrams, formulas and worked examples.

Workout selection | Nutrition & AI meal plans | Health rules | Dashboard scores | Coach & payments | API map

Built from the application source · 13 October 2026 · Every number in this guide is read from the code, not estimated. Second edition — matches the app as it works now.

START HERE

How to read this guide

Kotaana is made of three layers. Once you see which layer does what, every feature becomes easy to follow.

1

Screens

What you tap: dashboard, workout, nutrition, coach portal. Screens never decide anything — they ask the server and show the answer.

2

APIs

Small server addresses such as `/api/workout/generate`. Each one checks who you are, loads your data from the database, calls an engine, and saves the result.

3

Engines

The rules and maths: the workout engine, the nutrition engine, the health rules and the AI router. They hold the real "intelligence".

Words used in this guide

Word	Meaning in Kotaana
Dataset / library	The stored list of real exercises (1,002) and real foods (about 8,100 from USDA). The app only recommends things that exist here.
Verified	A flag on each exercise/food. The workout engine ignores anything not verified.
Filter (safety gate)	A hard yes/no test. If an exercise fails it, it is removed for that person — it can never come back through scoring.
Score	A number used to rank what is left (higher = better fit). Page 8 lists every point; page 12 shows how the final pick uses them.
Seed	A number made from your user ID + today's date. It makes the daily pick different each day but stable if you reload.
Readiness	Your 0–100 daily score from the check-in (energy, mood, soreness, sleep).
AI (DeepSeek / Gemini)	A language model called over the internet. It suggests; the app's own rules always have the last word.
Fallback	What the app does when something is missing (no AI key, no weather, no data) — it never invents numbers.

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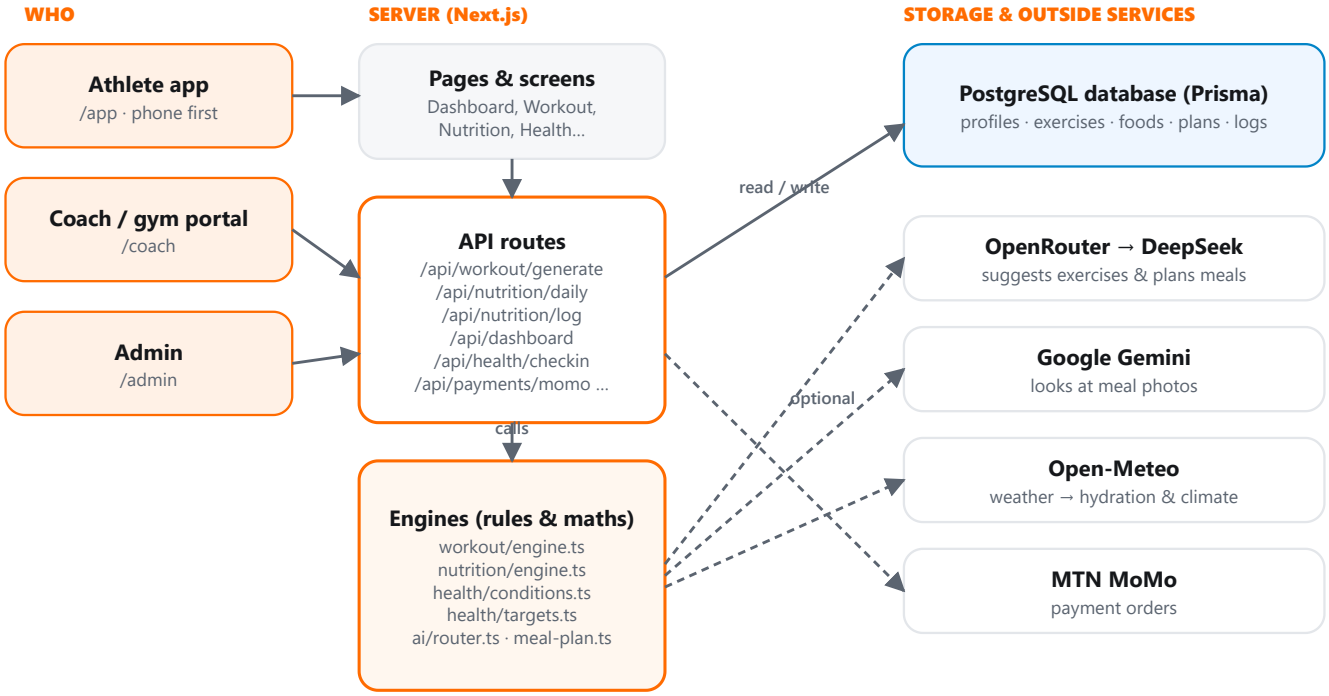
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PART A · SYSTEM

Architecture: who talks to whom

Every action travels the same road: screen → API route → engine → database, with outside services used only for extras.



Solid arrows = always. Dashed arrows = only when a key / network is available; otherwise the app falls back to its own rules.

The rule that keeps it safe

- **Screens never trust themselves.** Every API first calls `apiAuth(role)`: it reads your session and refuses the request if your role (athlete / coach / admin) is wrong.
- **Every row has an owner.** Reads are filtered by your user ID (or, for coaches, by an *active* coach–athlete link). That is how data reaches the right coach and nobody else.
- **AI is optional.** If no AI key exists, workouts and meals are still built by the app's own rules. If the AI answer is unsafe or unusable, it is thrown away.

API groups

Group	Routes	Used for
Workout	workout/generate · workout/start · workout/perform · workout/checkout · exercises/[id]	Build today's session after the questions, start it, record each block, check out, show how-to demos
Nutrition	nutrition/daily · nutrition/log · hydration	Plan meals, log/verify them, count water
Health	health/checkin · health/situation · health/followup · health/injury · health/document · ai/risk	Readiness, sick/injured cards, recovery follow-ups, documents, risk signal
Dashboard & reports	dashboard · questionnaire · ai/review · reports/overview · reports/series	Numbers, fitness status, weekly/monthly questions, the 3–7 day review
Coach & money	connections/search · connections/request · coach/* · payments/momo · payments/momo/approve	Pick a coach, share data, pay and confirm
Account	auth/* · profile · location · notifications	Sign-in (email or Google), profile, GPS, inbox

PART A · SYSTEM

A day in the life of one athlete

Follow one person through a normal day and watch which APIs and engines fire.



The API routes each step touches are written under it. Nothing is built when you log in.

Step	What you do	What the server does	Saved in
1	Sign in with email or Google	Checks password (bcrypt) or the Google token, makes a session. Athletes without a paid subscription get a free trial counted from sign-up; after that the app sends them to the payment page	users, accounts, profiles
2	Answer the check-in card	Readiness = **50 + energy×8 + mood×5 − soreness×9 + min(12, sleep)×2**, clamped 0–100 (Okay preset 3, 2, 3 and 7 h sleep = **85**). Saying “sick” or “injured” opens a second card (page 17)	daily_checkins, injuries, health_updates
3	Open the dashboard	dashboard collects the numbers, fitness status, notifications and any due review; nutrition/daily finds or builds today's meals. **No workout is built yet**	nutrition_plans (cache)
4	Answer the workout questions	Indoor/outdoor, how hard, aim, minutes. The answers are saved to your profile and **then** the workout is built (at most 3 builds a day). A serious illness or injury returns a rest day instead	profiles, workout_plans, usage_counters
5	Press Start	workout/start stamps the start time	workout_logs
6	Mark each block done (easy / ok / hard) or skip	workout/perform stores effort, updates the completed list, adds an energy estimate. The phone vibrates and/or beeps; timed blocks have a countdown that beeps at zero	exercise_performances, workout_logs, activity_data
7	Check out	Effort 1–10, how you feel, any pain and where. Pain creates an injury record so tomorrow avoids it	workout_logs, injuries
8	Eat and log meals (photo optional)	nutrition/log adds calories and macros and remembers the foods; a photo is checked by Gemini (max 10 a day)	meal_logs, usage_counters

What changes the plan after it is built: if you check in again or report an injury *after* today's plan was made, the workout screen shows “Your health info changed — rebuild”. Rebuilding counts towards the 3-a-day limit. Nothing rebuilds silently.

PART B · WORKOUT ALGORITHM

Step 1 — What the engine knows about you

Before choosing anything, `workout/generate` gathers everything relevant. It only runs after today's check-in exists. Nothing is guessed: if a value is empty the engine uses a stated default and records it.

Input	Comes from	How it is used
Goal, track, level, minutes	profiles (+ the 4 questions)	Session length, number of exercises, sets/reps, bonus points. "How hard" shifts level: easy −1, hard +1 (children never step up). A general goal with a weight target is read as that direction: ≥2 kg to lose → fat loss, ≥2 kg to gain → muscle gain
Equipment you have / excluded	profiles + your gym's equipment list (if you have a coach)	Hard filter: an exercise needing equipment you lack is removed
Age band	profiles (from date of birth)	Children: no barbell / cable / machine / sled work, never "hard", max 3 main blocks, 40 minutes, always general-fitness aim
Injuries & limitations	injuries table (active/recovering) + profile notes	Hard filter: knee, shoulder, back, wrist, ankle rules (page 6)
Health conditions, pregnancy, BMI, stress	profiles	Removes exercise types, caps effort, adds notes (page 7). Pregnancy from week 14 also removes lying-flat moves and crunches. BMI ≥ 30 removes high-impact moves. Stress ≥ 4 lowers readiness by 10
Readiness and feeling	today's daily_checkins (required)	Lowers effort and volume when low. "Unwell / a bit off" caps readiness at 40 and effort at easy
Last 14 days of performances	exercise_performances	Recent muscles/patterns (small penalty); last effort → progression. Exercises done in the last 3 days may repeat — in a new format (page 10)
This week's muscles & patterns	exercise_performances (7 days)	Variety scoring
Missed sessions	workout_logs (7 days) vs your weekly target	Adds a note: volume held conservative if 2+ missed
Training days, outdoor OK	profiles	Picks one outdoor day per week (seeded)
Seed	hash(user ID + today's date)	Daily rotation that stays stable on reload

Where the exercises come from

Dataset	Records
free-exercise-db	745
bryllim / workout-guide	256
mohamedatef	1
Total in the library	**1,002**

Each record has: name, body part, target muscle, secondary muscles, equipment, difficulty, steps, GIF/image, and safety tags (contraindications, age limits, role tags such as *stretch*, *warmup*, *core*, *outdoor*).

Honest data-quality note. The curation report marks all 1,002 records **"partial"** quality and many lack written instructions with "human safety review required before activation". The engine therefore filters by tags and keywords. A person reviewing the tags will make every filter on the next pages more reliable.

PART B · WORKOUT ALGORITHM

Step 2 — The safety funnel

Hard yes/no gates run first. An exercise that fails any gate is gone for that person today, whatever its score would have been.



Shape only — the real size of each step differs for every person. The app shows the final count as "candidateCount".

Gate	Exact rule in the code	Example
Verified	exercise.verified must not be false	Unreviewed imports are ignored
Equipment	"body only / none / bodyweight" always passes. Otherwise the exercise's equipment text must match something you own. Anything in your "not available" list is removed	You own only dumbbells → barbell exercises vanish
Difficulty	exercise difficulty rank (1–3) ≤ your level rank + 1	A beginner can see beginner and intermediate, never advanced
Age	Youth bands drop exercises tagged adult_only / high_risk and anything needing barbell, cable, machine, smith, sled or trap bar. Unknown age drops age_restricted / high_risk	Under-14 → bodyweight, bands, light dumbbells only
Injury	If injury status is unknown, drop high_risk / injury_sensitive / advanced_load tags. If known, each injury maps body part → forbidden tags & words (below)	Knee injury → jump, pistol, deep squat, box jump removed
Limitations & today's extras	"no jumping / no running / no impact" in your notes removes jump, run, sprint, impact. The build also adds today's avoid-words: lying-flat moves in later pregnancy, high-impact moves at BMI ≥ 30	
Health condition	Words listed by each condition (page 7) are searched in the exercise name, muscles, tags	Osteoporosis → crunch, twist, jump removed

Injury map (body part → what is removed)

Injury text contains	Contraindication tags	Words removed
knee	knee, squat_depth, impact	jump, pistol, deep squat, lunge jump, box jump
shoulder / rotator	shoulder, overhead	overhead press, snatch, handstand, dip, muscle up
back / lumbar	lower back, hinge	good morning, heavy deadlift, sit up, hanging leg raise
wrist	wrist, loaded wrist	handstand, planche, push up
ankle	ankle, impact	jump, sprint, box jump

If **nothing** survives the funnel, the engine stops with an error that lists what blocked it (equipment, age, level, injury…). It will not build an unsafe fallback — that is by design.

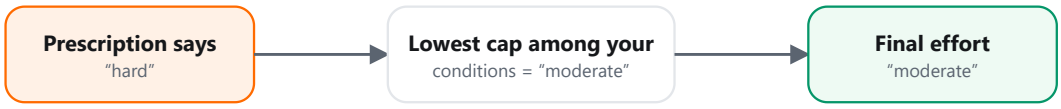
PART B · WORKOUT ALGORITHM

Step 3 — Health conditions change the rules

Each condition you tick in Health adds its own rule set. The same rules also remove foods (see the nutrition part).

Condition	Exercises removed	Effort cap	Foods removed
High blood pressure	sprint, burpee, box/jump/tuck jump, plyo, handstand, headstand	moderate	salty / processed
Heart disease	same as above	moderate	salty / processed
Diabetes (type 1, 2)	—	—	sugary foods
Asthma	sprint	—	—
Arthritis · Obesity	high-impact (jump, sprint, burpee, plyo, skipping)	—	—
Osteoporosis	high-impact + sit-ups, crunches, toe touch, twists	—	—
Chronic back pain	crunches, sit-ups, deadlift, hanging leg raise, superman	—	—
Sickle cell	high-intensity moves	moderate	+500 ml water
Epilepsy	handstand, headstand, rope climb	—	—
Kidney disease	—	moderate	salty foods
Pregnancy	high-impact, spinal flexion, twists, hanging, inversions (+ lying flat, crunches from week 14)	moderate	raw fish, liver, high-mercury fish, alcohol
BMI 30 or more (automatic)	jump, sprint, burpee, plyo, skipping	—	—
Feeling unwell today	—	easy, readiness ≤ 40	—
Child or teenager	heavy equipment, max-effort work	moderate	growth-friendly meals, no calorie restriction

How an effort cap works



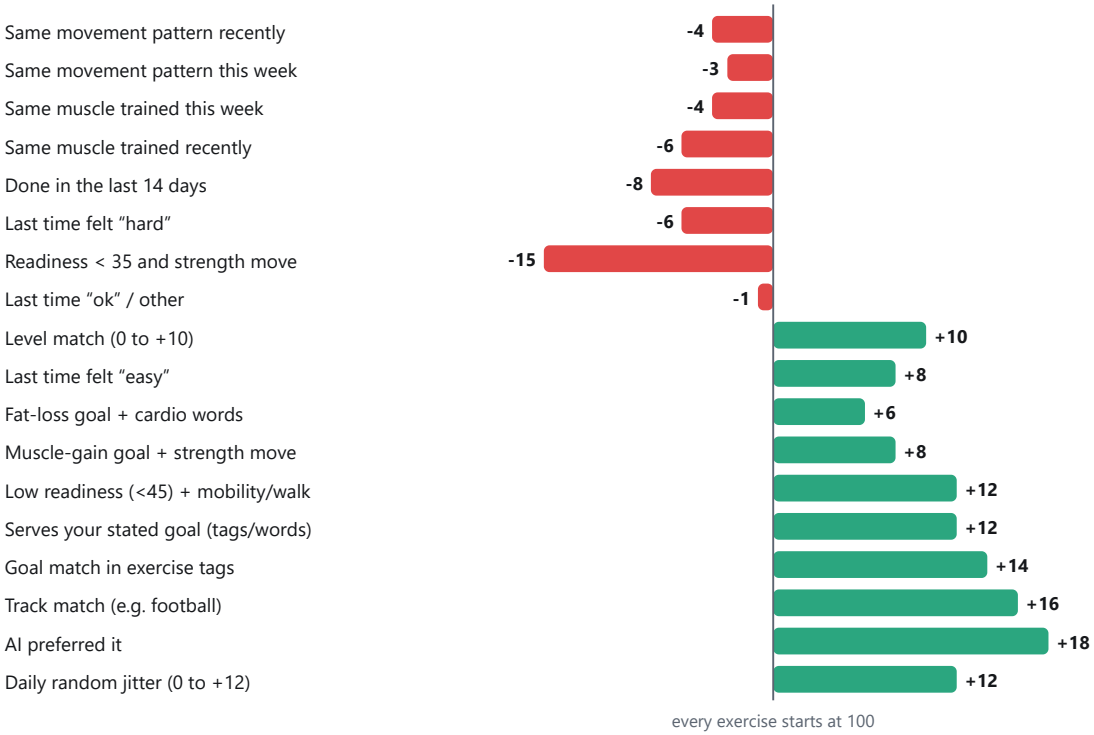
Caps only ever lower effort, never raise it.

Every active condition also adds a plain-language note that appears on the workout (“Adjusted for your health profile”) and a nutrition note on the meals page. These are conservative filters and reminders — they never replace a doctor's clearance.

PART B · WORKOUT ALGORITHM

Step 4 — Scoring: how “good for you today” is calculated

After the funnel, each remaining exercise gets a score that starts at 100 and moves up or down. Blocked exercises get −∞ (never chosen).



Signal	Why it exists
Recent use, muscles, patterns	Only small penalties now: repeating an exercise is allowed when it serves your goal (then it comes back in a new format)
Goal / track match	Fat loss favours cardio, muscle gain favours strength moves, endurance favours aerobic work, football track favours football-tagged moves
Readiness	Tired days favour mobility and walking and penalise heavy strength work
Last performance	Moves that felt easy are more welcome; moves that felt hard are less so
AI preferred	A DeepSeek suggestion adds +18 (page 11)

The scores now matter. Page 12 explains how the final pick uses them: the best-scored exercises form a short list and the daily seed chooses among them.

PART B · WORKOUT ALGORITHM

Step 5 — Building the session

The eligible exercises are sorted into role pools, then one session is assembled in a fixed order.

1. Role pools

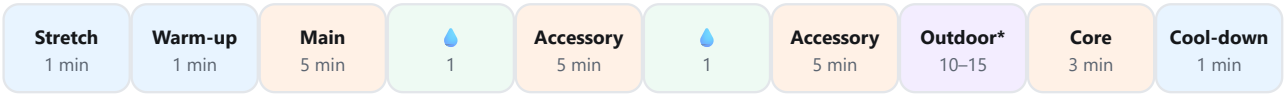
Pool	An exercise belongs when its tags/words say…
Stretch	role “stretch”, or stretch / mobility / flexibility / child pose / cat cow / 90 90
Warm-up	role “warmup”, or warm up / dynamic / march / jog / jumping jack / arm circle / leg swing
Core	role “core”, or core / abs / plank / dead bug / bird dog / oblique
Outdoor	role “outdoor”, or run / jog / walk / hike / sprint / hill
Main	everything left over (strength, cardio, bodyweight work). If your aim is fat loss, muscle gain or endurance, the main pool is narrowed to exercises that serve that aim — whenever at least twice as many as needed exist

If the stretch, warm-up or main pool is empty the engine stops and tells you which pool is missing — it will not make up a session.

2. How many main blocks

Your minutes	Main + accessory blocks
50 or more	5
38–49	4
28–37	3
under 28	2
Children (any time)	never more than 3

3. Order of the session



* only on your seeded outdoor day each week, when outdoor training is allowed and outdoor exercises passed the filters

Shown for 35 min; the number of main/accessory blocks changes with your minutes.

4. Keep it inside your time

Minutes are added up from each block's estimate. While the total is more than **your minutes + 5**, the engine removes accessory blocks one by one (never the first main block).

5. Save

The old plan for today is switched off (`isActive=false`), the new plan is saved with a version number and a reason, and the screen shows it. The session carries a stamp of what the engine used: defaults applied, candidate count, adjustment note and the policy line.

PART B · WORKOUT ALGORITHM

Step 6 — Sets, reps, progression and repeats

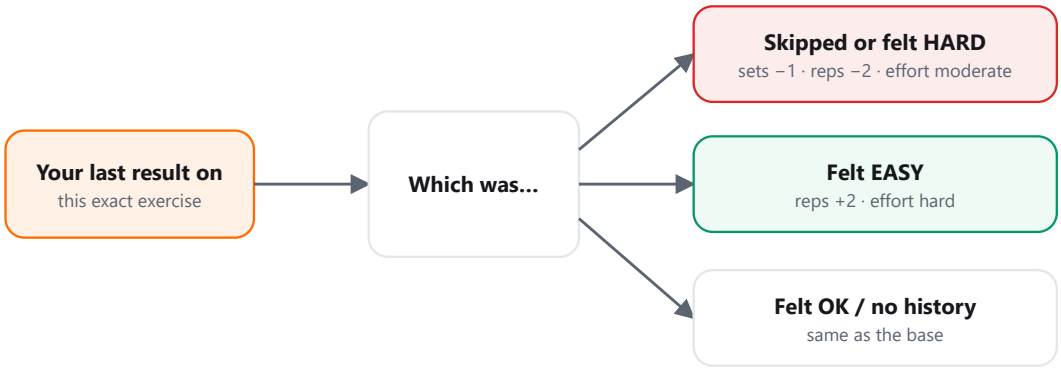
Choosing the exercise and deciding the dose are two separate jobs. The dose reacts to how your last attempt went.

Base dose by role

Role	Beginner	Intermediate	Advanced
Main / accessory	3 × 10 reps	3 × 12 reps	4 × 8–12 reps
Core	2 × 25 s	3 × 35 s	3 × 35 s
Outdoor	10 min	15 min	15 min
Stretch · warm-up · cool-down	45–60 s each, short rest	same	same

Rest is 60 s between main sets, 30 s for core. Minimum 2 sets and 6 reps after adjustments.

Progression from your last result



Repeating an exercise on purpose

If an exercise was done in the last 3 days and is chosen again…	Format given
Tempo 3-1-3	3 s down, 1 s pause, 3 s up — reps −3
Pause reps	2 s hold at the hardest point — reps −2
Pyramid	8-10-12 reps across the sets
Short-rest circuit	30 s rest, reps −1

The format is picked from the exercise and the day, shown on the block as “New format: …”, and explained in its note. The muscles keep adapting; the exercise still serves the goal.

Readiness changes the dose

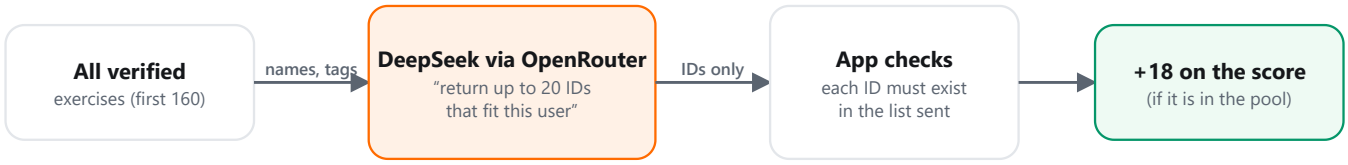
Readiness	Effect on the dose
≥ 45	Dose as above
< 45	Core holds are 25 s and easy; outdoor work is easy
< 35	Main sets −1 (minimum 2) and effort set to easy

Missed sessions: 2+ missed this week → note “volume held conservative before rebuilding frequency”; 1 missed → “normal volume with conservative progression”. This is a message about the plan's intent; the dose itself is governed by the table above.

PART B · WORKOUT ALGORITHM

Step 7 — What the AI does (and cannot do)

The AI is a suggestion box with strict rules. It never sees medical data as a decision-maker and cannot add exercises.



Question	Answer
When is it called?	On every workout build, only if an AI key exists (OPENROUTER_API_KEY, else GEMINI_API_KEY)
What does it receive?	Age band, level, track, goal, minutes, readiness, and for up to 160 exercises: id, name, equipment, category, muscles, tags, difficulty
What can it return?	A JSON list of at most 20 exercise IDs. Anything not in the list it was given is discarded
System instruction	"You may only choose from verified IDs… Never invent exercise IDs, medical diagnoses, or unsafe instructions."
Can it override safety?	No. Filters run on all exercises regardless of the AI. The AI line only adds points to exercises that already passed
If the call fails?	The workout is built without it — nothing breaks. One limit remains: it is shown the first 160 exercises of the library, not the person's safe pool, so it can only steer part of the choice
Which model?	OPENROUTER_MODEL, default deepseek/deepseek-chat-v3.1 (temperature 0.1)

PART B · WORKOUT ALGORITHM

Step 8 — How the final pick works now

An earlier edition of this guide showed that the scores were calculated but ignored when choosing. That is fixed and covered by tests.

What the code does

1. Eligible exercises are scored and sorted best-first into role pools.
2. For the main blocks the pool is first narrowed to your aim (if enough match).
3. `pickTop(pool, count, seed)` keeps the **best-scored short list** — the larger of 3× the number needed or the top 25% of the pool — and lets the daily seed choose among them.
4. So goal, readiness, recent work and the AI's +18 decide *who is on the short list*; the seed decides the day's rotation.

Tests that prove it

Test	Result
Aim = muscle gain, 20 muscle-gain + 20 neutral exercises, 6 days	Every main block is a muscle-gain exercise
AI prefers 5 of 40 exercises, 30 days	Chosen more often than without the AI
Same user, same day	Identical session every time
Recent exercise chosen again	Always carries a new format and a note
Child, advanced level, barbell available	No barbell, never hard, ≤ 3 mains
Pregnancy extras, knee injury, unwell	Crunches / jumps removed, effort easy

In plain words

- **Safety first:** filters run before any scoring, so unsafe exercises can never be picked.
- **Goal first:** if you want fat loss you do not get a strength-only day.
- **Stable day, fresh week:** the seed makes the plan the same all day and different tomorrow.
- **Repeats are allowed** when they help, never as an identical copy.

Still true: exercise data quality (all 1,002 records rated “partial”, many without written steps) limits how reliable the tag-based filters are, and the AI is shown only the first 160 exercises. Both are listed on page 22.

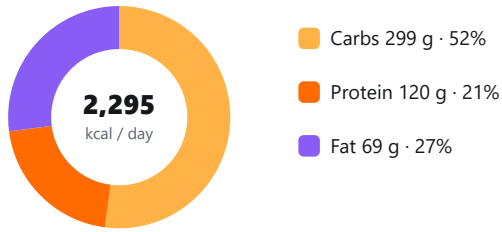
PART C · NUTRITION ALGORITHM

Step 1 — Daily targets and water

Before a single food is picked, the app works out how much you should eat and drink. This is plain arithmetic on your profile.

Calorie and macro targets (health/targets.ts)

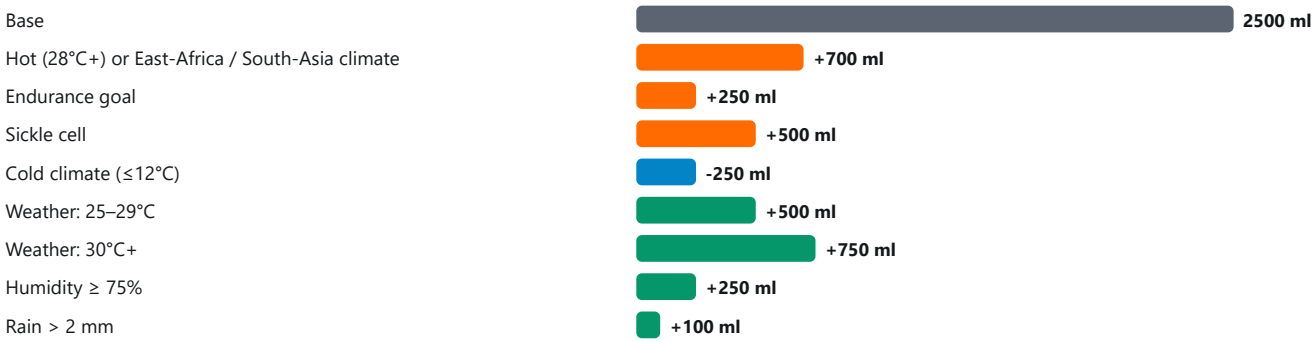
1. If you typed your own calorie target, use it.
2. Otherwise, **Mifflin-St Jeor**: `10×kg + 6.25×cm − 5×age + 5` (male) / `− 161` (female) / `− 78` (unspecified).
3. × activity factor `min(1.55, 1.2 + 0.05 × training days)`.
4. Goal: fat loss −400, muscle gain +300; at least 1,200 kcal.
5. Protein = **1.6 g/kg** (2.0 for fat loss or muscle gain).
6. Fat = **27%** of calories ÷ 9. Carbs = the rest ÷ 4.



Worked example — male, 30 y, 75 kg, 175 cm, 3 training days, general fitness: base = 750 + 1,093.75 − 150 + 5 = 1,698.75 → × 1.35 = 2,293 → rounded to **2,295 kcal**. Protein 75 × 1.6 = **120 g**. Fat 2,295 × 0.27 ÷ 9 = **69 g**. Carbs (2,295 − 480 − 621) ÷ 4 = **299 g**. If height, weight or age is missing and you set no target, the app falls back to 2,200 kcal / 130 g protein.

Children and teenagers (under 18) never get an adult goal adjustment or a deficit. Calories come from age: up to 8 y 1,600 · 9–13 y 1,950 (girls) / 2,200 (boys) · 14–17 y 2,200 girls / 2,600 boys / 2,400 otherwise, never below 1,400. Protein ≈ 1 g per kg, fat 30% of calories, carbs the rest. The meal plan adds dairy or alternatives and iron- and calcium-rich foods and gives no weight-loss advice.

Water target (ml per day)

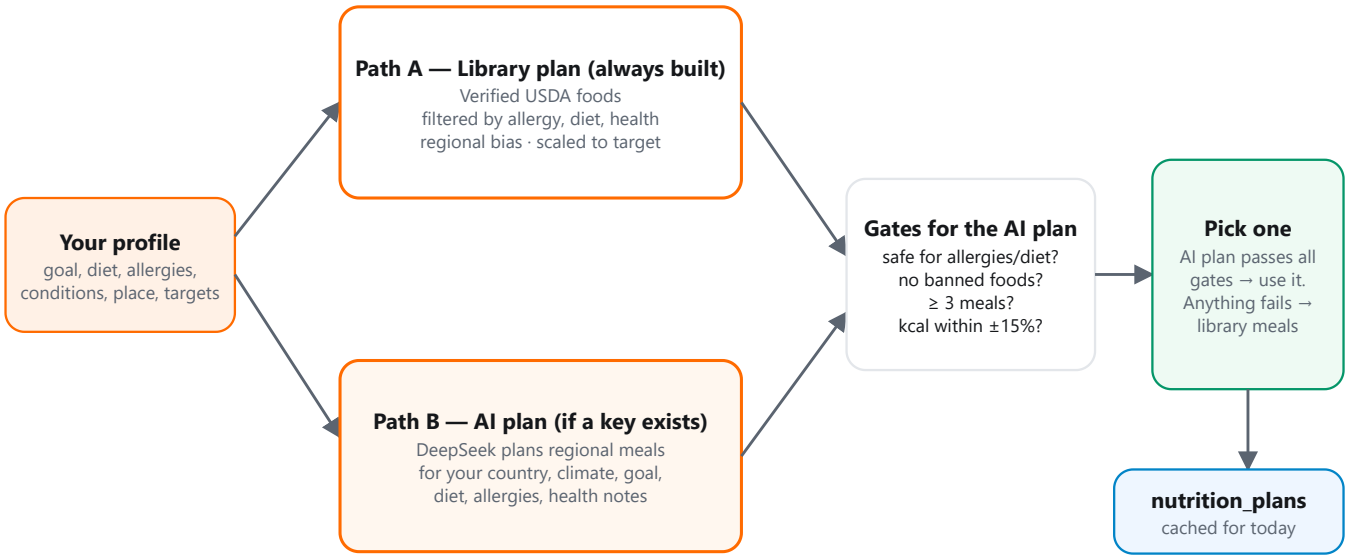


Open-Meteo gives the temperature, humidity and rain at your saved location. The total is clamped between 1,800 and 5,000 ml and shown as glasses of 300 ml. Without a location the base is used.

PART C · NUTRITION ALGORITHM

Step 2 — Two ways to plan, one safe result

The library plan is built first and is always ready. The AI plan is offered second and only used if it passes every gate.



Detail	How it works
Cache	One plan per person per day (table nutrition_plans). Reloading never re-calls the AI. The plan is rebuilt only if something that feeds it changes: goal, diet, allergies, calories, weight, height, meals/day, cultural rules, conditions, pregnancy, location or training days. The “New meal plan” button forces a rebuild
What the library always supplies	Hydration target, health notes, location label and the fallback meals — even when the AI wins
Which meals	Breakfast, lunch, dinner (+ a snack if you chose 4 meals) and a post-workout meal on days you have a workout plan
Learns what you eat	Meals you log (and what the photo check sees) are remembered. Foods you log at least twice in 30 days are favourites: the library puts them first and the AI is told to use some of them when they suit the targets
Fruit	One or two of the day's meals or snacks include fruit (never every meal), chosen from real fruit in the library — juices, dried fruit, jams and tomatoes are excluded. The AI is asked for the same, using fruit that is local and in season
Limits	Rebuilding the whole plan: 5 a day. The first plan of the day is free
Why two paths	The library is reliable but limited to what USDA lists; the AI knows local dishes (ugali, injera, dal, tortilla…) the dataset lacks. Safety is guaranteed by running the same allergy / diet / health checks on both

PART C · NUTRITION ALGORITHM

Step 3 — The library algorithm in detail

How the app builds meals from real USDA foods (about 340 Foundation + 7,793 SR Legacy records).

#	Stage	What happens
1	Safe pool	Keep foods that are verified, have calories, are safe for your allergies (milk, peanut, tree nut, egg, soy, fish, shellfish, gluten and tag matches) and your diet (vegan / vegetarian), and do not match a health ban (salty, sugary, raw fish…)
2	Region	Country text → east_africa / south_asia / europe / americas / global. If at least 4 safe foods match that region's staples (e.g. plantain, cassava, ugali, teff, beans, lentils, rice, sweet potato for East Africa), they are moved to the front of the pool
3	Climate	Weather ≥ 28°C = hot, ≤ 12°C = cold; with no weather, East-Africa / South-Asia count as hot. Hot raises the water target
4	Pick by role	From the pool, take 3 protein foods (chicken, egg, yogurt, tuna, lentil, bean…), 3 carbs (rice, oat, potato, banana…), 3 vegetables and 2 fats (avocado, nuts, olive oil…). Position in each list = (daily seed + offset) mod list size, so it rotates daily
5	Assemble	Breakfast = protein + carb (+ fruit) · Lunch = protein + carb + vegetable · Post-workout = protein + carb (+ fruit) · Dinner = protein + vegetable + fat. Fruit goes to breakfast, post-workout or both, rotating daily
6	Per-food numbers	Each food's calories/macros per 100 g × its serving grams ÷ 100
7	Scale to target	factor = target ÷ plan total, limited to 0.75–1.8, and applied to every food. Then meal totals are recomputed. (In a test the plan reached about 88% of a 2,380 kcal target; the AI path or a snack is needed to close the rest)

Limits you should know. The scale factor is capped, so with a high target the day can still end below it. Foods are chosen by name patterns, not a nutritionist's rules, so combinations can be plain. This is exactly why the AI path exists.

Health rules in nutrition

The same condition table from page 7 supplies the food bans and notes (high blood pressure/heart/kidney → no salty foods; diabetes → no sugary foods; pregnancy → no raw fish, liver, high-mercury fish, alcohol). They are applied to the library pool *and* to every food name the AI proposes.

PART C · NUTRITION ALGORITHM

Step 4 — The AI plan, logging and photo check

How DeepSeek is asked, how its answer is policed, and how your meals are counted.

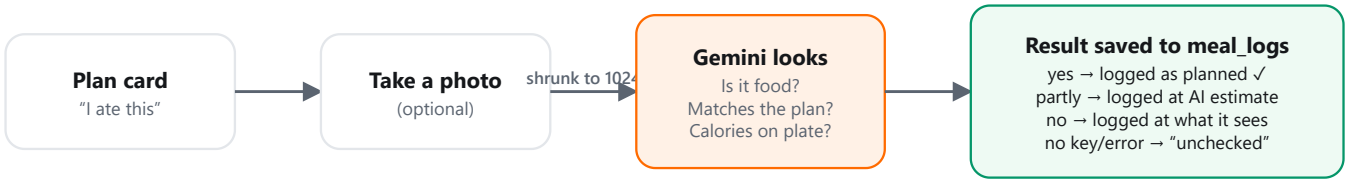
The question sent to DeepSeek

Sent	Details
Place & climate	City, country, hot / mild / cold
Goal & targets	Daily calories, protein, carbs, fat
Safety	Diet style, allergies, cultural / religious rules, health notes
Format	Exact slots in order, 2–5 foods per meal, portion + calories + macros per food, total within 5% of target, a day-seed so it varies
Role	"Registered-dietitian style planner… local, affordable foods… respect every allergy… no medical diagnoses". Temperature 0.6

How the answer is policed

1. Must be valid JSON with a meals list; numbers are clamped to sane ranges.
2. **Any** food that fails the allergy, diet or health-ban check cancels the whole AI plan (it is never quietly edited).
3. At least 3 meals, and the total must be between 85% and 115% of your calorie target.
4. If any gate fails the library meals are used. The screen says which source built today's plan: "✦ AI regional plan" or "Verified food library".

Logging a meal and the photo check

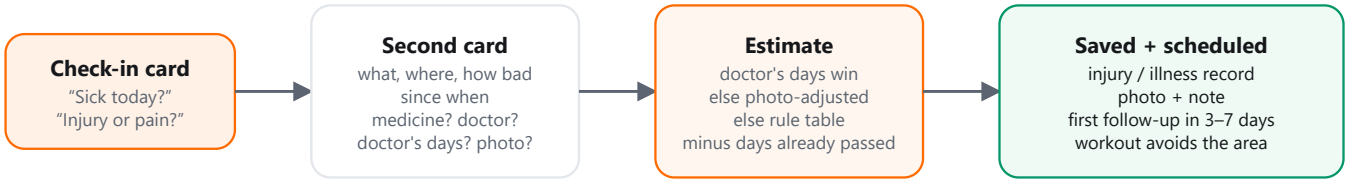


Honest limits. AI calorie and macro numbers are *estimates* and are labelled so. Photo checking needs a Gemini key; without one the meal is logged as planned and marked unchecked. Photos are shrunk and analysed, not stored. Meals you log feed the dashboard's calories, macros and nutrition score.

PART D · HEALTH & RULES

Sickness, injury and recovery

Every check-in asks whether you are sick or hurt. Yes opens a second card; the answers set how long to full fitness and how the workouts change meanwhile.



Recovery estimate (days) — health/recovery.ts

Severity (1–5)	Injury base	Illness base
1	5	2
2	12	4
3	28	7
4	56	10
5	90	14

× a type factor: fracture 2.2 · dislocation 1.8 · tendon/ligament tear 1.7 · sprain/strain 1 · wound/burn 0.9 · bruise 0.6 · flu/malaria/pneumonia 1.6 · stomach 0.8. A doctor's number replaces all of it. A photo can scale the estimate between 0.7× and 1.4×.

Follow-ups: every 3, 5 or 7 days

Situation	Next follow-up in
Getting worse, or severity 4–5	3 days
Severity 3	5 days
Severity 1–2 and stable	7 days

At each follow-up the athlete says better / same / worse, rates pain 0–10, says if still on medicine, may enter a new doctor timeline and may add a photo. "Better" shortens the remaining time by 20%; "worse" lengthens it by 30% + 2 days. Healed percent = days passed ÷ estimate (±), and the injury is marked recovered when the athlete says so or it is better, pain 0 and 60% of the time has passed.

Fit for workout? — health/fitness.ts

State	When	What the app does
Fit for workout	no open injury or illness, not pregnant	Normal plan
Train lightly	any open injury/illness (severity under the limits), or pregnancy	Plan avoids the area, effort capped; shows days to full fitness or week/trimester
Rest for now	illness severity 3+ or injury severity 4+	No workout is built; gentle walking and stretching only

Photos: a vision model comments cautiously ("looks like it is healing", "consider seeing a doctor — spreading redness") and may nudge the estimate. It never diagnoses; the app shows "estimates only — not a diagnosis" everywhere. Photos are visible to the athlete's current coach and the administrator.

PART D · HEALTH & RULES

Reviews, questionnaires, limits and access

How the app keeps learning about you, and the fair-use rules that protect it.

The 3–7 day review (a larger model)

Part	How it works
When	Due 3 days after the last review (needs at least 2 check-ins). Written automatically the first time the dashboard opens after it is due, once a day at most
Input	Everything recorded in the period: check-ins, workout days, effort and pain, meals and calories vs target, water, weight, open injuries and their progress, latest questionnaire answers, pregnancy, age group
Model	OPENROUTER_REVIEW_MODEL (default deepseek/deepseek-r1), then the standard model, then plain rules if neither answers. It is told to use only the numbers given, never diagnose, never advise weight loss for children
Output	Summary, what is working, what to watch, up to 5 suggested changes — shown on the dashboard, sent as a notification and visible to the coach

Questionnaires that keep your data current

Questionnaire	When	What it does
Weekly situation	every 7 days (after day 6)	Weight, sleep, stress, pain, goal feeling, regular meals, pregnancy feeling → saved for the review; weight and stress update your profile
Monthly update	every 30 days (after day 28)	Weight, target weight, training days, level, goal, minutes, new health issue → updates the profile so every plan uses today's facts, then a monthly review is written

Fair-use limits (per person per day)

What	Limit	What happens at the limit
Workout builds	3	Current plan stays; message says to try tomorrow
Meal photo checks	10	Meals can still be logged without a photo
Meal-plan rebuilds	5	Today's plan stays
Water, check-ins, weight, meal logs, follow-ups	unlimited	—

Access: trial and subscription

State	Meaning
Subscription active	30 days from the coach's or admin's confirmation (or an admin grant)
Free trial	7 days from sign-up (admin can change it)
Expired	Workout, nutrition and dashboard APIs answer 402 and screens redirect to the payment page. Settings, notifications, onboarding and the payment page stay open
Paywall off	Admin switch: everyone has access

PART E · THE REST

Dashboard: where every number comes from

One call (/api/dashboard) reads your recorded data and returns the lot. When nothing is recorded the value is empty and the screen says so — no invented scores.

Number	Formula	Data used
BMI	kg ÷ (height in m)^2. <18.5 underweight · <25 healthy · <30 overweight · else obese	profile weight, height
Goal %	(current − first recorded weight) ÷ (target − first recorded weight), clamped 0–100	weight_history, target weight
Calories today	sum of calories of meals you logged vs your daily target	meal_logs
Macros today	sum of protein/carbs/fat of logged meals vs targets	meal_logs
Adherence (7 days)	days with a finished workout ÷ your weekly sessions, max 100%	workout_logs
Streak	consecutive days (counting back from today, or yesterday) with a finished workout	workout_logs
Fitness score	100 × (0.5 × 30-day consistency + 0.3 × average completion + 0.2 × BMI band). BMI band: healthy 1, overweight 0.6, obese 0.3, unknown 0.6. Empty until you finish a workout	workout_logs, BMI
Recovery	today's readiness, else average of the last 7 days	daily_checkins
Fitness status & days to fitness	from open injuries/illness and pregnancy (page 17); days = the longest remaining time	injuries, profile
Body fat label	ACE bands, by gender (e.g. men: <6 essential, <14 athlete, <18 fitness, <25 average)	profile
Weeks to goal	|target − weight| ÷ 0.4 kg per week, limited to 4–52	profile
Calories burned	5 MET × kg × hours for each finished block (duration, or sets × 50 s) plus tracked runs — an estimate	activity_data, run_activities

Health overview score (0–100)

Part	Excellent / Good / Fair / Poor	Score
Recovery	≥80 / ≥65 / ≥45 / lower	95 / 80 / 55 / 30
Sleep (7-day average hours)	≥7.5 / ≥7 / ≥6 / lower	95 / 80 / 55 / 30
Nutrition (7-day average kcal ÷ target, needs ≥3 logged days)	Good 85–115% · Fair 70–130% · else Poor	85 / 55 / 30
Hydration (7-day, needs ≥3 days)	≥90% / ≥70% / ≥50% / lower	95 / 80 / 55 / 30
Overtraining	Low / Moderate / High: sessions vs weekly plan, and sore (≥4) on top of extra sessions	95 / 60 / 30

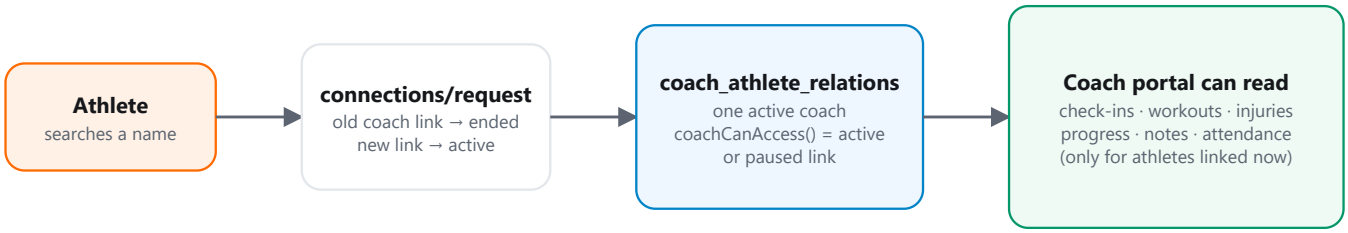
Overall = the average of the parts that have data (at least two). Parts with no data are shown as “No data” and a note says why (e.g. “Sparse meal logging”).

PART E · THE REST

Coach connection, capacity and payments

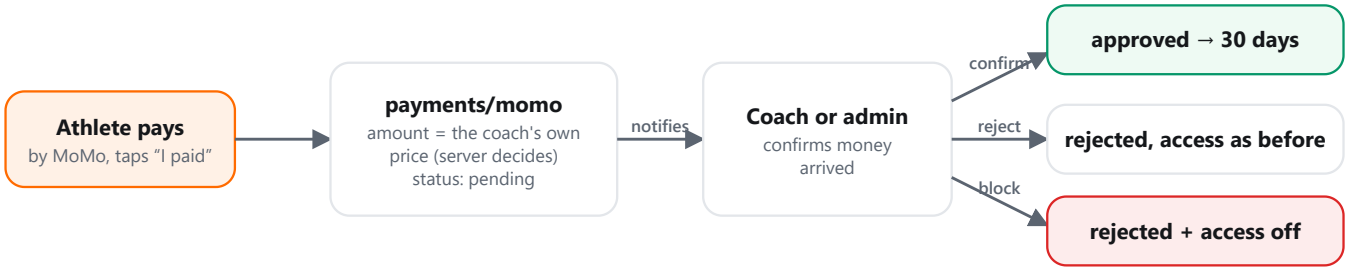
Data follows the athlete's current coach. Money follows the coach's own price. Coaches look after 70 members free.

Connecting



Leave or switch coach → the link ends and the old coach's access stops immediately. Admin can see all athletes and coaches.

Paying



Rule	Detail
Fixed price	A coach sets price (RWF per 30 days) and MoMo number in Settings. The payment API ignores any amount from the browser for coach memberships
30-day cap	Every approval or direct grant sets expiry to now + 30 days; renewing early never stacks days
Who can confirm	The coach who owns the order (and only for athletes still linked to them) or an admin. An answered order cannot be answered again
No coach	The athlete pays Kotaana directly (platform price, default $10 converted to RWF) and an admin confirms
Community	Coaches register as an individual coach, a gym or a sports club. Members see "You're part of the [name] gym community" on the dashboard and menu
Capacity	Free up to 70 members. 71–150: $20 · 151–230: $30 · 231–310: $40 · then +$10 for each further 80, per 30 days. When a coach is full, athletes cannot choose them and the coach is notified once a day. The upgrade is a payment order priced by the server; an admin confirms it and the new limit lasts 30 days, then returns to 70 for new joiners

PART D · THE REST

API map

Every route, what it needs and which engine/table it uses. “A” = athlete, “C” = coach, “Ad” = admin.

Route	Who	Does	Engine / tables
GET workout/generate	A	Today's session; auto-builds when missing or health changed	workout engine, ai rerank · workout_plans
POST workout/generate	A	Rebuild now (after the questions)	same
POST workout/start · POST workout/checkout	A	Stamp the start; save effort, feeling, pain at the end	workout_logs, injuries
POST workout/perform	A	Save effort per block, update completion, add calories	exercise_performances, workout_logs, activity_data
GET exercises/[id]	A	Steps, demo media for “How to do it”	exercises
GET / POST nutrition/daily	A	Today's meals (cached) / force a new plan	nutrition engine, meal-plan AI, targets · nutrition_plans, foods
POST nutrition/log	A	Log or undo a meal, optional photo check	Gemini vision · meal_logs
GET / POST hydration	A	Glasses of water for your local day	hydration_logs
GET / POST health/checkin	A	Daily readiness card	daily_checkins
POST health/situation · GET/POST health/followup	A	Sick / injured cards, recovery estimate, follow-ups with photos	injuries, health_updates, recovery.ts, photo AI
health/injury · health/document	A	Injuries and medical files that feed the safety filters	injuries, medical_documents
GET/POST questionnaire · GET/POST ai/review	A	Weekly / monthly answers; 3–7 day and monthly reviews	questionnaires, athlete_reports, review model
GET dashboard	A	All dashboard numbers	targets, many tables (read only)
reports/overview · reports/series · ai/weekly-summary · ai/monthly-progress · ai/risk	A	Recorded-data reports and risk signal	reports-data
GET connections/search	A	Find coaches by name (price, MoMo)	users, profiles
POST / DELETE connections/request	A	Choose a coach / leave	coach_athlete_relations, notifications
coach/members · coach/access · coach/notes · coach/follow-ups · coach/injuries · coach/sessions · coach/attendance	C	Manage members and coaching tools	relations + member data
PATCH coach/momo	C	Set price, MoMo number and name	profiles
GET admin/config · POST admin/subscriptions (paywall)	Ad	Which keys are set; paywall on/off and trial days	app_config
POST payments/momo	A, C	Create a payment order (amount decided by server)	payment_requests, notifications
POST payments/momo/approve	C, Ad	Confirm / reject / block	payment_requests, subscriptions
payments/momo/webhook	MoMo	Provider status updates (signed)	payment_requests
admin/metrics · admin/subscriptions · owner/platform-momo · platform/momo	Ad	Totals, price, access, platform code	app_config, subscriptions
profile · location · notifications · progress/photo · track/run	A	Profile, GPS & weather place, inbox, photos, runs	profiles, run_activities…
auth/*	all	Sign-up, sign-in (password / Google), reset	users, accounts

PART E · THE REST

Database map and honest limits

Which tables feed which decision — and where the system is weaker than it looks.

Decision	Reads	Writes
Build workout	profiles, exercises, injuries, exercise_performances, workout_logs, daily_checkins, coach profiles (equipment), usage_counters	workout_plans, usage_counters
Build nutrition	profiles, foods, meal_logs (favourites), workout_plans, Open-Meteo	nutrition_plans, profiles (water target)
Log food / water	nutrition_plans, hydration_logs	meal_logs, hydration_logs, usage_counters
Health follow-up	injuries, health_updates, profiles	injuries, health_updates, notifications
Reviews	everything recorded in the period + questionnaires	athlete_reports, questionnaires, profiles
Dashboard	profiles, weight_history, daily_checkins, meal_logs, hydration_logs, workout_logs, activity_data, injuries, athlete_reports	—
Coach view	coach_athlete_relations + the athlete's data above	notes, follow-ups, attendance, sessions
Access & money	subscriptions, payment_requests, app_config, profiles (capacity)	subscriptions, payment_requests, notifications, profiles

What to keep in mind

1. Exercise data quality. All 1,002 records are rated "partial"; many lack instructions and need a human safety review. Every filter depends on those tags.

2. The AI sees only the first 160 exercises of the library, not the person's safe pool, so it steers only part of the choice.

3. Library-only meals run low. Without an AI key a plan can end around 88% of the calorie target.

4. Estimates are estimates: AI calories/macros, photo checks, recovery days, calories burned (MET) and the fitness score formula are approximations and labelled so.

5. Not medical advice. Condition rules and recovery estimates are conservative aids, not a diagnosis or clearance from a doctor.

6. Photos are stored in the database today; move them to object storage (S3/R2) before many users. Children need a guardian-consent step for launch.

Quick reference — numbers worth remembering

Item	Value
Exercises / foods in library	1,002 · about 8,130
Main blocks by time	≥50 min 5 · ≥38 4 · ≥28 3 · else 2 (children ≤ 3)
AI exercise bonus	+18, and it now steers the short list
Daily limits	3 workout builds · 10 meal photos · 5 meal-plan rebuilds
Follow-ups	every 3, 5 or 7 days by seriousness
Reviews	3–7 days (large model) · monthly
Free coach capacity	70 members; $20 → 150, $30 → 230, +$10 per 80
Access	30 days, never stacked; 7-day trial

CHK:ok